

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF GEOLOGICAL ENGINEERING



GEOLOGICAL FIELD MAPPING EXERCISE AT PEMENASE

DATE: 9th to 13th September 2024

BATCH 3 GROUP4

Group Members	
Name	Index Number
Anthony Essel Prepeh	1748722
Alfred Baffour Asare	1747422
Erica Oteng Twum	1748622
Yanney John	1748922
Ofori Sarfo Kingsford	1748322
Osei Boateng Justice	1748522
Adams Fahim	1746622
Abena Biamah Sekyere	1748822
Kumi Mensah Doreen	1748022
Afriyie David Owusu	1746722

Table of Contents

Group Members	ii
CHAPTER 1	1
INTRODUCTION STATEMENTS	1
BACKGROUND REGIONAL GEOLOGY	1
LOCAL GEOLOGY	1
Relief and landforms	3
PROBLEM STATEMENT	5
MAIN AIM	6
ACKNOWLEDGMENT	7
CHAPTER 2	8
MATERIALS	8
METHODOLOGY	8
MAPS	9
TRAVERSE MAP	9
OUTCROP MAP	10
STRUCTURAL MAP	11
GEOLOGICAL MAP	12
FIELD WORK	13
3.1 DAY 1 FIELD WORK	13
FIRST OUTCROP FOR DAY 1	13
THE SECOND OUTCROP FOR DAY 1	15
THIRD OUTCROP FOR DAY 1	16
3.2 DAY 2 FIELD WORK	17
FIRST OUTCROP FOR DAY 2	17

THE SECOND OUTCROP FOR DAY 2.....	18
THE THIRD OUTCROP FOR DAY 2.....	19
FOURTH OUTCROP FOR DAY 2.....	20
FIFTH OUTCROP FOR DAY 2	21
3.3 DAY THREE FIELD WORK	22
FIRST OUTCROP FOR DAY 3.....	22
SECOND OUTCROP FOR DAY 3	23
THIRD OUTCROP FOR DAY 3	24
FOURTH OUTCROP FOR DAY 3.....	25
3.4 DAY 4 FIELD WORK	26
FIRST OUTCROP FOR DAY 4.....	26
SECOND OUTCROP FOR DAY 4	27
THIRD OUTCROP FOR DAY 4	28
FORUTH OUTCROP FOR DAY 4	29
3.5 DAY 5 FIELD WORK	30
THE ONLY OUTCROP FOR THE DAY	30
LABWORK.....	31
Thin Section Preparation	31
MICROSCOPIC ANALYSIS OF SOME OF THE THIN SECTIONS	32
Magnetite.....	32
Biotite	32
Muscovite	34
Feldspar	34
Quartz	36
UNDER PPL OBSERVATION.....	36

CHAPTER THREE	37
3.0 RESULTS AND DISCUSSIONS.....	37
Petrography of Metasedimentary rocks.....	37
Petrography of Metavolcanic rocks.....	39
Petrography of Granitoids	40
CHAPTER FOUR	44
4.0 CONCLUSION AND RESULTS.....	44
4.1 APENDICE	44
4.1 REFERENCES.....	45

LIST OF FIGURES

Figure 1 Base Map	4
Figure 2 Traverse Map	9
Figure 3 Outcrop Map	10
Figure 4 Structural Map	11
Figure 5 Geological Map	12
Figure 6 Shale	14
Figure 7 Metavolcanic rock	15
Figure 8 Granitoid	16
Figure 9 Metasedimentary rock	17
Figure 10 Metavolcanic rock	18
Figure 11 Quartzite.....	19
Figure 12 Phyllite.....	20
Figure 13 Meta-conglomerate	21
Figure 14 Slate	22
Figure 15 Slate	23
Figure 16 Metasedimentary rock	24
Figure 17 Metavolcanic rock	25
Figure 18 Metamorphosed claystone	26
Figure 19 Highly weathered Metasediment	27
Figure 20 Metamorphosed shale	28
Figure 21 Metasedimentary rock	29
Figure 22 Metasedimentary rock	30
Figure 23 Photomicrograph of schist in: (A) XPL, (B) PPL, both showing a dark mineral identified as Magnetite	32
Figure 24 Photomicrograph of a mineral in: (A) PPL, showing a brown color, (B) XPL, showing a dark color is identified as Biotite	33
Figure 25 Photomicrograph of a mineral in: (A) PPL, colorless, (B) XPL, showing rainbow colors is identified as Muscovite	34
Figure 26 XPL, mineral labelled "Or" is orthoclase feldspar	35
Figure 27 Photomicrograph of a mineral in: (A) PPL, colorless, (B) XPL, showing rainbow colors is identified as Quartz	36
Figure 28. Photomicrograph showing minerals of: (A) slate in XPL, (B) Meta-conglomerate in PPL, (C) Quartzite in PPL, and (D) Schist in PPL.	38
Figure 29 Photomicrograph of Metabasalt in XPL	39
Figure 30 Photomicrograph showing Granite in: (A) PPL, (B) XPL	40
Figure 31 Photomicrograph foliations in: (A) slate in PPL, showing “D1” and “S1” as deformation one and Foliation respectively in a quartz vein, (B) Slate in XPL, (C) series of foliations due to different deformation history in slate, and (D) metamorphosed shale in XPL	41
Figure 32 Evidence of folding	43
Figure 33 Traverse map used on field.....	44

CHAPTER 1

INTRODUCTION STATEMENTS

Geological Field mapping exercise is a very important exercise in geology, because geology is a field-oriented discipline, geologists do say 'the best geologist is the one who has seen most rocks. You can never be a great geologist until you've become a field orientated geologist. It is on this basis, this fieldwork was carried out, and a report is submitted to the Geological Engineering Department, on the geology of a section of the Bosomtwe area, Ashanti Region. This field exercise is also a requirement for GED 365 in the Geological Engineering Department, KNUST. The mapping exercise commenced for 2024/2025 academic year, on the 9th of September and ended on the 13th of September 2024. Comprising of three Batches; four groups each. Batch three, Group four worked around the Pemenase area. The field mapping exercise lasted for five days (5 days).

BACKGROUND REGIONAL GEOLOGY

This is a field mapping exercise done in Bosomtwe District in central part of Ashanti Region, Capital city; Kuntunase, Latitude 6° 24' South and 6° 43' North and Longitude 1° 15' East and 1° 46' West, Land surface area of 422km² and population density of 275.2 per km².

The district is underlain by Precambrian rocks of the Birimian and Tarkwaian formations. They are associated with granites and metamorphosed sediments of phyllite and schist. The Soil type in the district has developed over a wide range of highly weathered parent materials such as granite, conglomerate, sandstone, phyllite etc.

LOCAL GEOLOGY

For batch 3 group 4, between five to nine outcrops were encountered per day. Mostly metasedimentary rocks with quartz veins with appreciable number of igneous rocks, either slightly weathered or highly weathered. Observable geological structures were measured and described of which photographs were taken. Outcrops of similar composition and same rock type would not be repeated for a day's observation for the sake of this paper. A summarized note on descriptions will follow herein but photographs will be placed in an appendix with proper titles for easy reference.

Rock Exposure:

The land of traverse covers the first moderate hills rising from the farm plains to the river on the south side. The only major rock exposures are found on the base of the hills and in the

valleys of dried-up rivers. A few prominent outcrops are exposed by road cuts. Otherwise, the area is covered with farmlands and rivers and good forest soils develop over the entire area.

However, on the top of hills and on their immediate slopes, many boulders are found completely exposed or partially exposed; the size (several meters in diameter) and their angularity suggest the boulders have not undergone transportation and therefore suggest they reflect the sub-surface bedrock. Those at the base of the hills, on the other hand, show roundness and therefore it will be difficult to suggest they reflect the sub-surface bedrock.

Additionally, most of the flat lands has anthills which suggest the lithology of the below ground.

Rock Units:

In the Pemenase township and immediate surroundings the layered units are made of fine grain to medium size metasedimentary rocks; In two close locations, the relationship between two similar units could be estimated. Likewise, boundaries between two distinct units could be estimated but same cannot be said for their actual contact.

Attitude measurements of layered rock units suggest most of the units trend North –East with parallel foliations if applicable, with mostly two sets of fractures, one that run parallel to foliation planes and the other cuts across the foliation planes. (Fig 3. Metasedimentary rock outcrop)

A few others, like quartzite, had no foliations meta-conglomerates which seemed to have undergone too much transportation.

In valleys or gullies left by dried up rivers are slated looking rocks overlaid by topsoil.

volcanic rocks encountered are either slightly metamorphosed, partially exposed or boulders with evidence of transportation. Examples are roundedness in the large grains in a porphyritic volcanic rock mostly found at the base of hills; Some are nearly close to six feet and others weathered to the size of a cannon ball.

Intrusive rock units:

The main portion of intrusive materials in mostly metasedimentary rock units and volcanic units is quartz by composition, which are mostly in the form of veins and veinlets of variant thickness. Large quartz intrusions which can be traced over long distances are typical of metasediments in the Ninko area. Here, different formation of layering due to different deformation is observable too.

The next step observed locally in valleys is sediments are close to complete metamorphism.

Additionally, Metasedimentary rocks in Pemenase and Adumasa are intruded by a greasy looking material which causes the overlying layers to fall off easily. Composition is not one that is familiar yet, so a specimen was taken for lab observations.

Conditions of rock units:

In most instances intrusive rocks have been weathered and veinlets are so rearranged as to transform into a breccia rock.

No chemical alteration was observed on field, suspected region of alterations would be accounted for after laboratory observations.

Climate and Vegetation:

Pemenase was normal during the exercise, but the ground always had moisture in it and vegetation was always covered in drops of water every morning.

Human Geography:

Pemenase is mainly dominated by ‘Asante Twi’ speaking people with few other Akan dialects spotted among them. The major occupation of the inhabitants is mainly subsistence farming, and the major crops produced are

Cocoa, cassava, plantain, banana, orange and pineapples; others include vegetables and maize. Most of these farming activities are carried out in the valleys and hills which in most cases have loamy soils and within a region that has a high-water table.

Relief and landforms

The study area is a terrain with rugged relief. There are both low lying and high lying landforms. There are different landforms within the study area, these include deep and steep valleys.

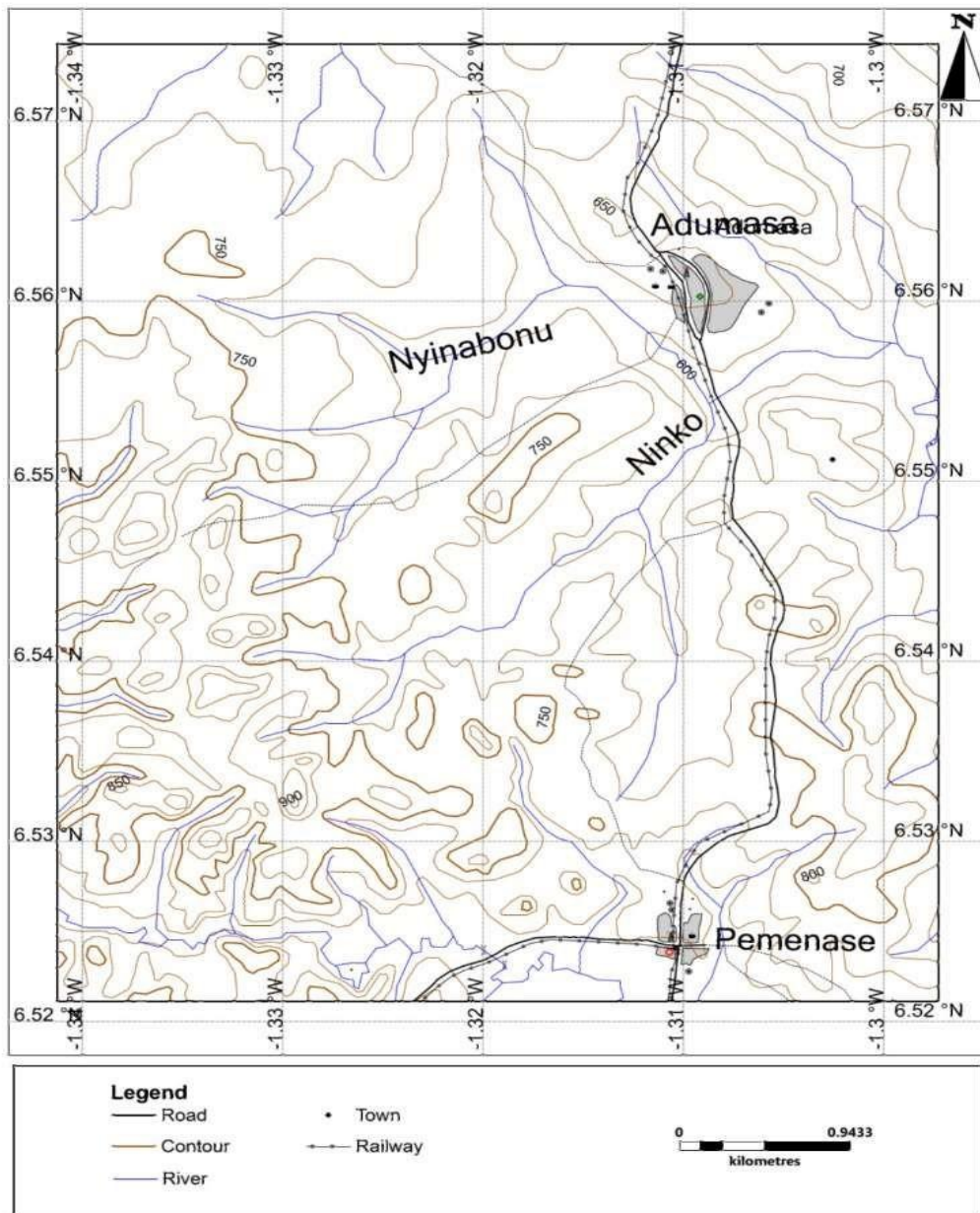


Figure 1 Base Map

PROBLEM STATEMENT

The field mapping exercise conducted in Bosomtwe of the Ashanti Region was aimed at studying and documenting the geological features of the area. Field mapping is an essential technique in geology that involves systematic observation, recording, and interpretation of surface and subsurface geological characteristics. This process helps geologists create detailed geological maps, which are crucial for understanding the distribution of rock types, geological structures, and the geological history of an area.

The Bosomtwe area, known for its unique geological setting, presents a variety of geological features, including sedimentary formations, igneous intrusions, and tectonic structures that provide insight into the area's tectonic history and environmental conditions. This report focuses on the methods used during the mapping exercise, the findings from the field, and the implications of the geological features observed in understanding the region's broader geological context.

MAIN AIM

The primary objective of this field mapping exercise is to produce a detailed geological map of the area and to interpret the significance of the rock formations and structures encountered. The information gathered from this exercise will contribute to the broader understanding of the geology of the Ashanti Region and its implications for natural resource exploration and environmental management.

ACKNOWLEDGMENT

All thanks to God almighty for the provision of good health throughout our stay on the main KNUST campus. Our profound gratitude to the members of the field committee for facilitating us in the field. And to all group members for the hard work and participation, we did our best as a team. God richly bless the local people of Pemenase for their hospitality.

CHAPTER 2

MATERIALS AND METHODOLOGY

MATERIALS

The materials we used include:

Hammers and Chisels

Compasses with Clinometers

Global Positioning System (GPS)

Hand lenses and Tapes

Field notebooks

Protractors, pencils and erasers

Acid bottles

Field clothing

METHODOLOGY

Field Technique and Data Collection

1. Identification of rock type
2. Structural Measurement
3. Sketches And Photographs
4. Weathering And Erosion Features

MAPS

TRAVERSE MAP

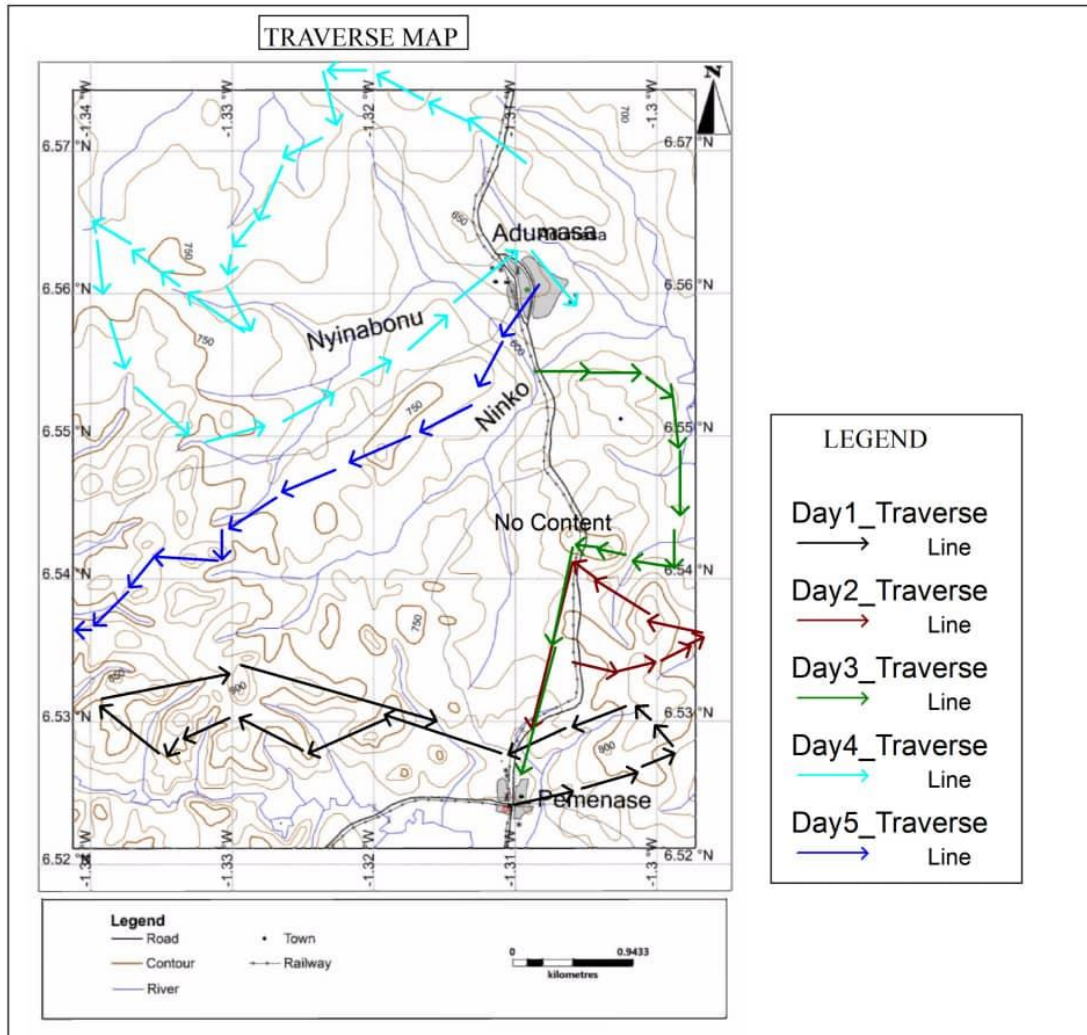


Figure 2 Traverse Map

OUTCROP MAP

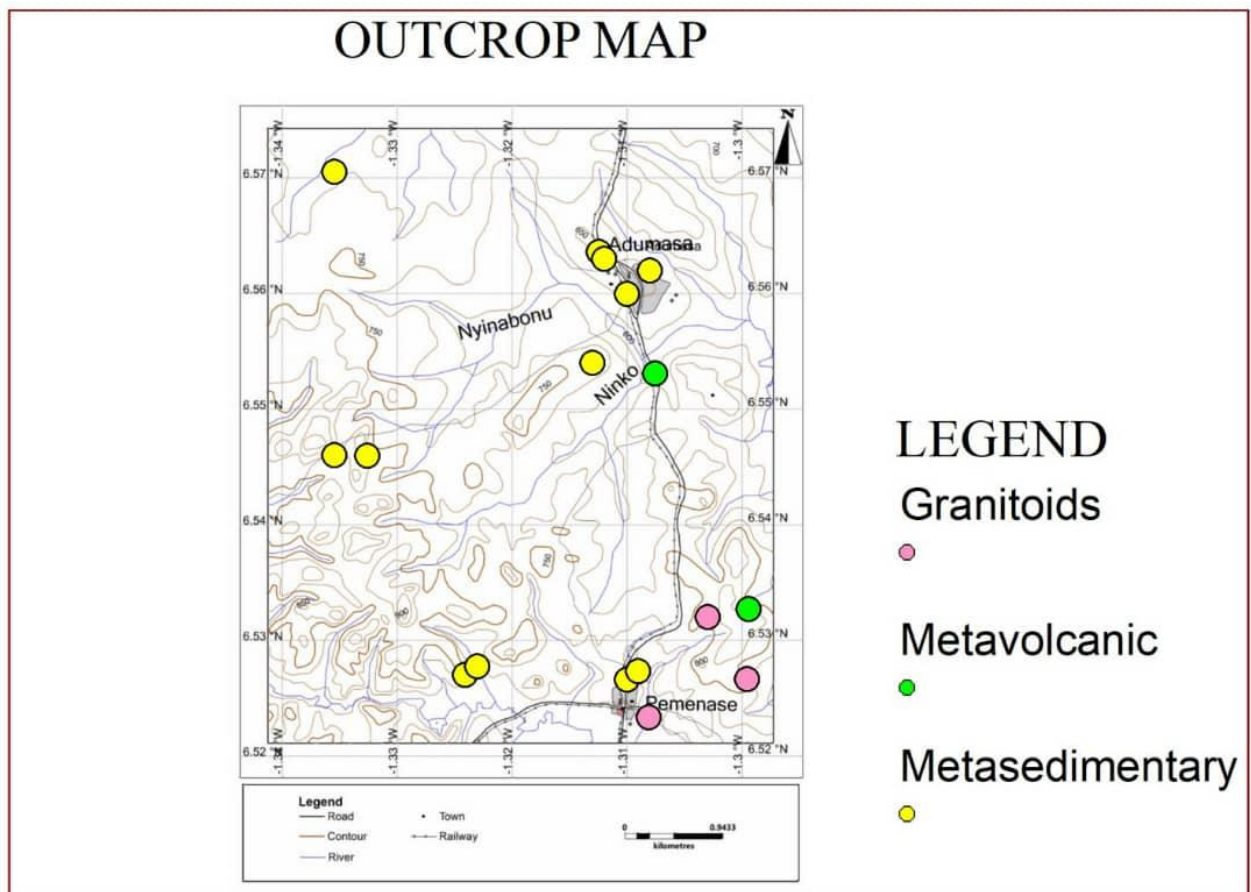


Figure 3 Outcrop Map

STRUCTURAL MAP

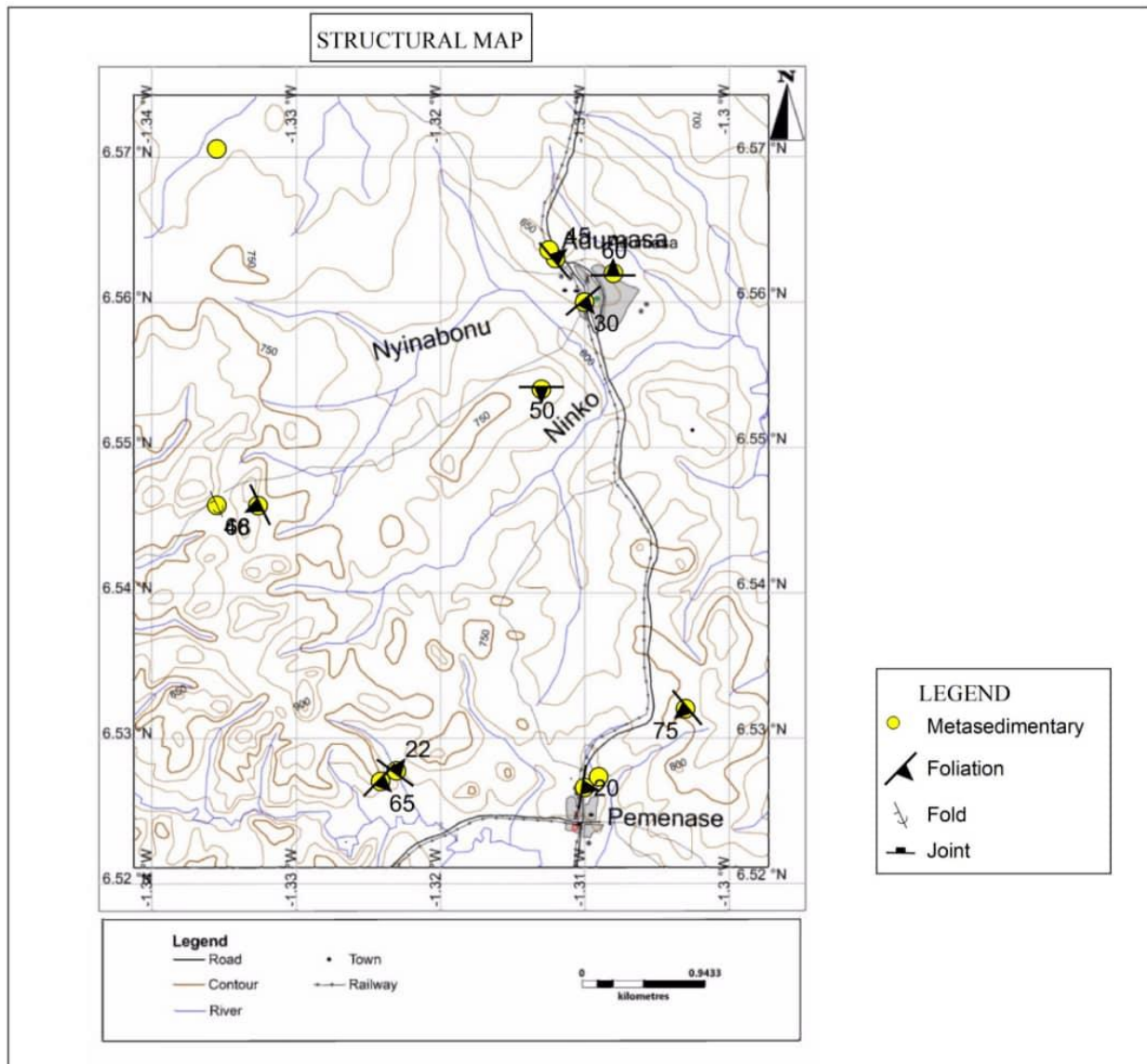


Figure 4 Structural Map

GEOLOGICAL MAP

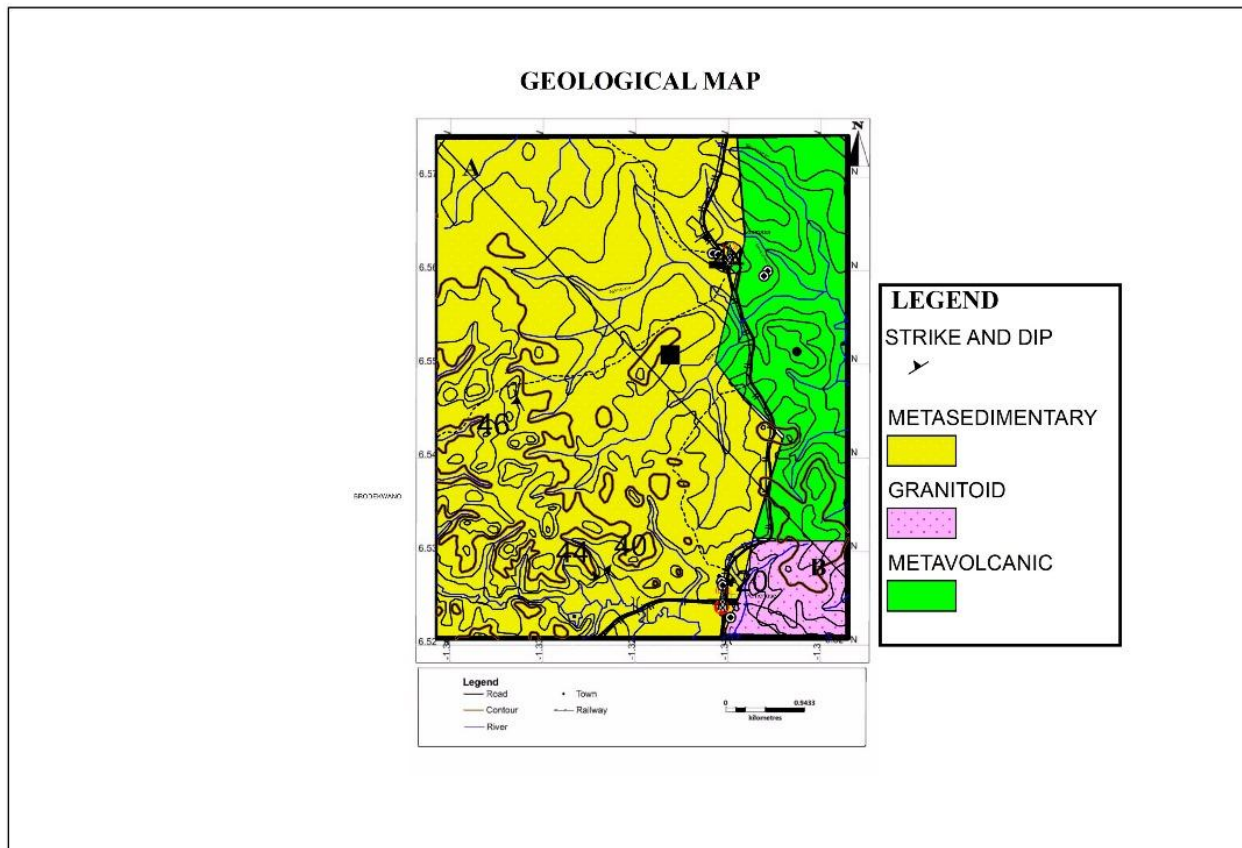


Figure 5 Geological Map

FIELD WORK

The exercise was carried out in batches of four groups, each group with close to eleven members with at least two facilitators for a day's trip. The duration for field traversing was close to about six hours per day. Students and their facilitators upon arrival at their start point along the roadside traverse by foot throw farm paths, bush tricycle trails or would have to create way for themselves in the bushes.

You don't go through any other way that leads outside the map even if that is the only way you see where you now find yourself. You solely rely on your GPS device to know your current position, your base map and compass to know

where you are heading, your machete to create way, your first aid kit for minor injuries, your common sense to avoid major injuries and be aware of your surroundings and your eyes to identify and describe outcrops when you are lucky to find one.

Note date, general location and weather conditions about the day and the beginning and ending time of observation. But most importantly, record factual data about each outcrop you encounter at each location and take enough samples for lab observation. This is very important because that is the reason for all the walking.

3.1 DAY 1 FIELD WORK

Date: 9th September 2024

Time: 8:32am

Location: Pemenase Police station

GPS coordinates; N 06.524825 W 001.310077

Elevation: 199m 1.1

FIRST OUTCROP FOR DAY 1

The field mapping exercise at Pemenase began at coordinates 6.524837°N, 1.310699°W. An exposed outcrop located 97 meters from the T-junction near the police station, we identified a foliated metasediments including shale, siltstone, mudstone, and claystone. The shale exhibited a reddish-brown color, fine-grained mica flakes, and a weathered dark green appearance. The outcrop's structure showed a strike of 041° and cross-cutting fractures and veins parallel to its layers, which varied in thickness. The outcrop also displayed jointing (discontinuities) with a joint plane striking at 015°, a dip of 52°, and a direction of S75°E. The stratification of the layers suggests an undisturbed depositional history, where younger layers rest above older ones. The outcrop was medium- to fine grained, jointed, and classified as

meta-sediment due to signs of deformation and metamorphism. The picture below is the outcrop.



Figure 6 Shale

THE SECOND OUTCROP FOR DAY 1

The second outcrop was a metavolcanic rock, characterized by a porphyritic texture with both fine- and coarse-grained materials. A further analysis revealed the presence of clasts embedded in a fine matrix with veinlets, suggesting volcanic activity followed by deposition and subsequent weathering processes. Weathering was evident in the dark brown coloration and moderate surface degradation. The picture below is the outcrop.



Figure 7 Metavolcanic rock

THIRD OUTCROP FOR DAY 1

The third outcrop was located at 6.5207°N, 1.3008°W and was poorly exposed. It was characterized by a medium to coarse-grained texture, with angular grain shapes and interlocking grains. Only the strike of the bedding planes was measured, with a value of 028°. The picture below is the outcrop



Figure 8 Granitoid

3.2 DAY 2 FIELD WORK

Date: 10th September 2024

Time: 8:45am Location: Pemenase mosque

GPS coordinates; N 06.52577 W 001.30718 Elevation: 204m

FIRST OUTCROP FOR DAY 2

The second day's traverse followed a roadside with well-exposed outcrops, revealing a metamorphic rock predominantly composed of 80% quartz, identified as metasedimentary rock. The fresh surface of the outcrop was brown, while the weathered surface had a greyish ash color. It exhibited a medium- to fine-grained texture, with angular grain shapes and interlocking grains.

The rock showed foliation, where layers resembled sedimentary structures. The first foliation plane (S1) and associated deformation (D1) were observed to be parallel to the original bedding plane (S0). Attitudes were measured for the later foliation planes:

- S2: strike 078°, dip direction 178°, dip angle 79°
- S3: strike 002°, dip direction 282°, dip angle 62°.

These measurements suggest significant deformation during rock's metamorphic history. The picture below is the outcrop



Figure 9 Metasedimentary rock

THE SECOND OUTCROP FOR DAY 2

The next outcrop was located at 6.5207°N, 1.3008°W and was partially exposed. It was identified as a metavolcanic rock with a fine-grained, porphyritic texture. Quartz veins were the predominant structures, and the outcrop displayed a dark fresh surface. The picture below is the outcrop.



Figure 10 Metavolcanic rock

THE THIRD OUTCROP FOR DAY 2

The next outcrop was located at 6.5287°N, 1.30035°W and was partially exposed. It consisted of transported rock, and as a result, no attitude measurements were taken. The outcrop was identified as Quartzite, a metasedimentary rock. The picture below is the outcrop.



Figure 11 Quartzite

FOURTH OUTCROP FOR DAY 2

We encountered our next outcrop which was located at 6.52978°N, 1.32828°W, situated at the base of a hill. The outcrop was partially exposed and not scattered. It was moderately weathered, with a weathered color of dark green and a fresh color of grey ash. The grain size ranged from fine to medium, with no interlocking grains. Attitudes were measured with a strike of 013°, a dip direction of 154°, and a dip angle of 39°. The outcrop was described as a metamorphosed shale, probably schist or phyllite.



Figure 12 Phyllite

FIFTH OUTCROP FOR DAY 2

The next outcrop was located at 6.52837°N, 1.32658°W. It was partially exposed and scattered, with a moderate degree of weather. The weathered surface was greenish brown, while the fresh surface was gray ash in color. The outcrop had a medium to coarse grain size, mostly angular pebbles with a fine ground mass. The rock showed signs of metamorphism, but no attitude measurements were taken since it was classified as a transported rock. The picture below is a metamorphosed conglomerate.



Figure 13 Meta-conglomerate

3.3 DAY THREE FIELD WORK

Date: 11th September 2024

Time: 8:45am Location: Roadside

GPS coordinates; N 06.55308, W 001.30756

Elevation: 208m

FIRST OUTCROP FOR DAY 3

The first outcrop encountered on the third day was located at 6.54362°N, 1.30268°W. It was a highly weathered metamorphic rock, with a weathered color of yellowish brown and dark green and a fresh surface of reddish brown. The rock exhibited fractures (jointing), and although no attitude measurements were taken, the dip was recorded to assess the outcrop's deviation from normal. It had a fine grain size, interlocking, well-sorted grains. The outcrop was massive, scattered, and had undergone metamorphism. The mineral composition included small amounts of quartz and many feldspars. It was said to be transported.



Figure 14 Slate

SECOND OUTCROP FOR DAY 3

The second outcrop encountered on the third day was located at 654202°N and 1.30497°W. It was well exposed, with a weathered color of reddish brown. The rock exhibited foliations that moved in groups. The grains were interlocked and characterized by a fine to medium grain size with angular shapes. The outcrop was identified as slate.



Figure 15 Slate

THIRD OUTCROP FOR DAY 3

The next outcrop in the field was located at coordinates 6.55512°N and 1.31398°W. It was not well exposed and exhibited high weathering, characterized by a weathered color of reddish brown. The outcrop contained quartz veins and occurred in layers. It was composed of fine to medium grains with an angular grain size that were not interlocking. The attitude of the veins was measured, with a strike of 024°.



Figure 16 Metasedimentary rock

FOURTH OUTCROP FOR DAY 3

The next outcrop in the field was located at coordinates 6.52773°N and 1.32303°W. It was well exposed with mature fractures and had a weathered color of gray and a fresh color of white. The outcrop was coarse-grained, with interlocking grains. It had undergone metamorphism, featuring multiple quartz intrusions of varying widths, some as narrow as 1 mm, which were aligned parallel to the rock layers and present in the joints. The attitude of the quartz vein was measured, with a strike of 024°.



Figure 17 Metavolcanic rock

3.4 DAY 4 FIELD WORK

Date: 12th September 2024

Time: 8:32am Location: Adumasa

GPS coordinates; N 06.56040 W 001.30884

Elevation: 199m

FIRST OUTCROP FOR DAY 4

The first outcrop we identified on the fourth day was located at 6.52946°N and 1.30947°W. It was highly weathered, with a reddish brown to black weathered color. The outcrop was found along a footpath, occurring in layers with visible fractures and foliations, indicating metamorphic terrain. The outcrop was predominantly fine-grained (90%) and exhibited moderate weathering. The foliations were aligned with the north, striking at 210° and dipping at an angle of 51° with a dip direction of S304°E.



Figure 18 Metamorphosed claystone

SECOND OUTCROP FOR DAY 4

The next outcrop in the field, situated at coordinates 6.53288°N and 1.29725°W, was well exposed and showed signs of moderate weathering with a reddish-brown weathered surface. A quartz vein was observed within the joints, and the rock exhibited a coarse-grained texture.



Figure 19 Highly weathered Metasediment

THIRD OUTCROP FOR DAY 4

We found the next outcrop, situated at coordinates 6.54599°N and 1.33264°W, was partially exposed and showed signs of moderate weathering, with a dark to brown weathered color. It had a fine-grained texture and a strike of 154°.



Figure 20 Metamorphosed shale

FORUTH OUTCROP FOR DAY 4

The next outcrop, situated at 6.54610°N and 1.33548°W, was well exposed and severely weathered, featuring a reddish-brown to dark brown weathered surface. Folds were present in the rock, which had a medium to coarse-grained texture. The structural measurements recorded a strike of 155°, a dip direction of 265°, and a dip angle of 2



Figure 21 Metasedimentary rock

3.5 DAY 5 FIELD WORK

Date: 13th September 2024

Time: 8:38am Location: Adumasa

GPS coordinates; N 06.56968 W 001.31082

Elevation: 195m

THE ONLY OUTCROP FOR THE DAY

This was the only outcrop we encountered along the roadside. It was highly weathered, with a dark brown to black weathered surface and appeared in layers. The outcrop had fine-grained, well-sorted material with non-interlocking grains. The strike of the bed was 228° , with a dip angle of 88° and a dip direction of $S107^{\circ}E$.



Figure 22 Metasedimentary rock

LABWORK

Thin Section Preparation

Over 22 rocks were sampled from the field of which sixteen (16) representative rocks were used to prepare thin sections at the Geological Engineering Laboratory of KNUST. Rock was first cut using the Rock Cutting machine. A particular face of the rock sample to be analyzed was selected and smoothened. The smoothening begun with the roughest surface (P60) of the abrasive papers through P80, P120, P180, P240, P400, P600 and then the finest surface using P1000. The glass slide of size 24mm by 48mm on which the specimen was to be bonded was roughened using a slurry made of a mixture of silicon powder 220 and water. The slide was then thoroughly cleaned and made to dry. A mixture of Epoxy and Hardener in an appropriate ratio was prepared and used to bond the prepared smoothened face of the rock specimen to the roughened surface of the glass slide and the bond was left for 168 hours (7 days) for reasons of lab availability but about 48 hours can do the job to properly dry and harden. The bonded slide was placed on the cutting edge of the cutting and polishing machine to cut off the unwanted part of the rock specimen and the surface trimmed and polished using the Grinding machine. The prepared thin section was finalized to a size of 30µm. Observations were made under the polarizing microscope to identify minerals as well as the geological structures using the 4x and 5x objective lens magnification.

The higher magnification (40x) was used when observing the interference figures of the minerals. All structural characteristics and findings during this study were used to identify the rock type were dealing with, and this aided in giving the rocks names other than the field names.

MICROSCOPIC ANALYSIS OF SOME OF THE THIN SECTIONS

Magnetite

Mineral appears as an opaque, black to dark mineral, and it doesn't transmit light. There is no cleavage in minerals.

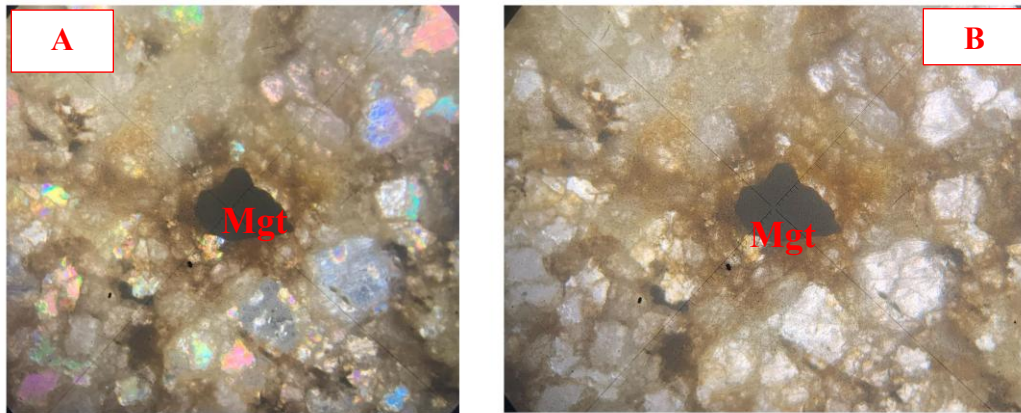


Figure 23 Photomicrograph of schist in: (A) XPL, (B) PPL, both showing a dark mineral identified as Magnetite

Biotite

PPL

Color: Brown

Relief: Moderate positive

Habit/Form: Euhedral crystals found as tabular grains with irregular outline. Grains are bent, particularly in slate (fig. 28. A)

Pleochroism is intense in granite but less intense in the slate. The reason is that absorption is stronger when the cleavage traces are parallel to the vibration plane of the lower Nicol.

Cleavage is in one direction and parallel to each other.

XPL

Anisotropic mineral

Interference color: 2nd order to 3rd bright

Extinction: 3 degrees

Twinning: No twinning was observed for this

Uniaxial (-)

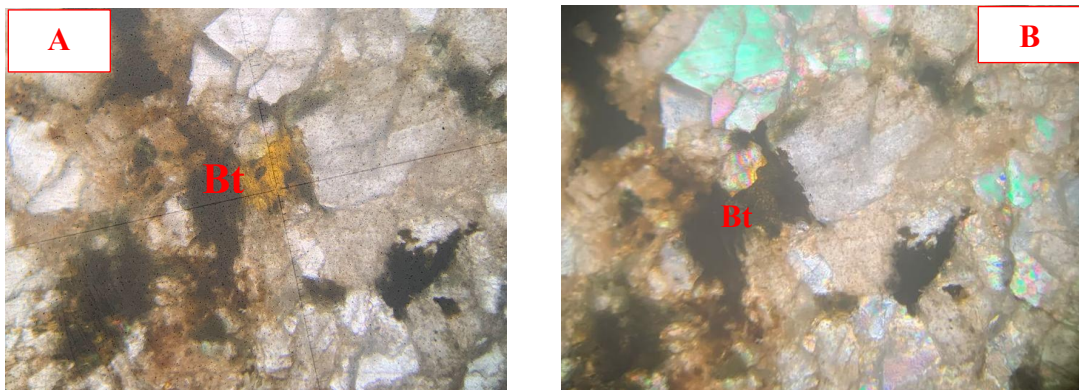


Figure 24 Photomicrograph of a mineral in: (A) PPL, showing a brown color, (B) XPL, showing a dark color is identified as Biotite

Muscovite

PPL

Color: Colorless

Relief: Moderate positive

Habit/Form: Well-formed tabular crystals,

Pleochroism: absent

Cleavages: Very perfect in one direction

XPL

Anisotropic mineral

Interference color: 2nd order to 3rd brilliant

Extinction: 0 degrees

Twinning: No twinning was observed for this

Uniaxial (-)

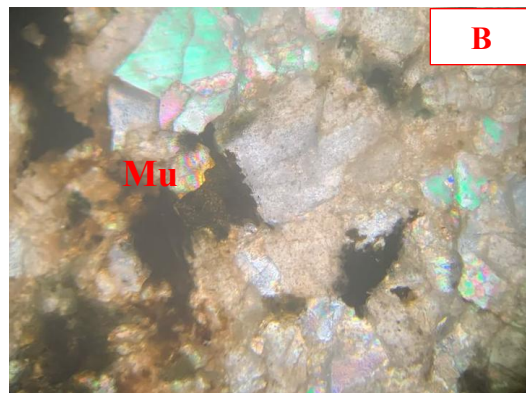
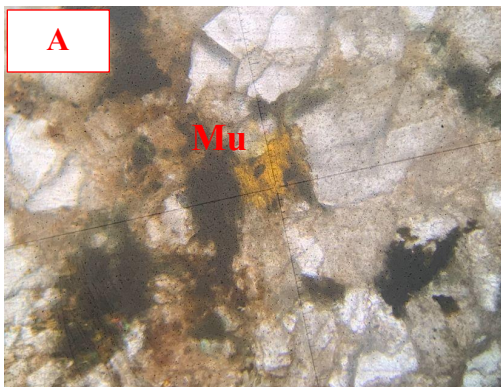


Figure 25 Photomicrograph of a mineral in: (A) PPL, colorless, (B) XPL, showing rainbow colors is identified as Muscovite

Feldspar (Orthoclase)

PPL

Color: Colorless

Relief: Low negative

Habit/Form: anhedral grains in igneous rocks. Crystal is elongated and roughly tabular.

Cleavage: Difficult to see due to the low relief as compared to the epoxy resin.

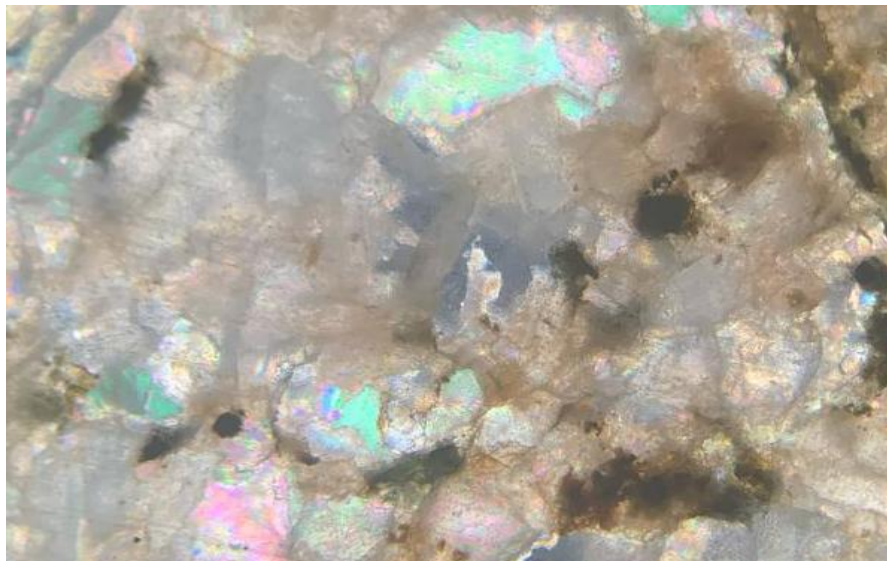
XLP

Anisotropic mineral

Interference color: 1st order (gray to white)

Extinction angle: 12 degrees

Twins: Carlsbad twinning



XPL

Figure 26 XPL, mineral labelled "Or" is orthoclase feldspar

Quartz

PPL

Relief: Low positive

Habit/Form: Anhedral and granular in the granite (fig. 30) but irregular in slate. (fig. 28. A)

Color: Colorless

XPL

Anisotropic mineral

Interference color: 1st order white to gray

Extinction: Undulous in the slate (Fig. 28. A)

Twins: Absent

Uniaxial (+)

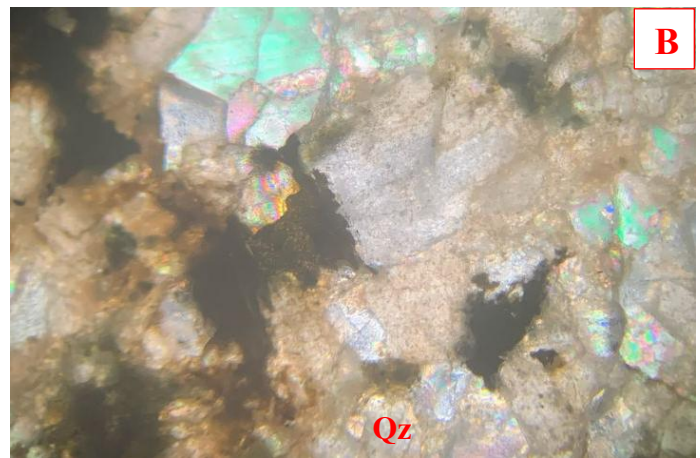
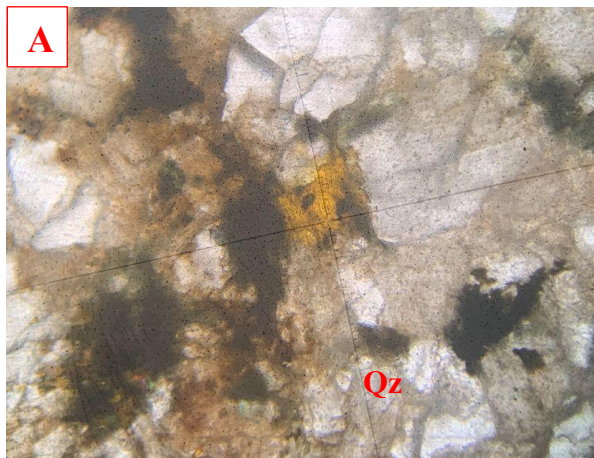


Figure 27 Photomicrograph of a mineral in: (A) PPL, colorless, (B) XPL, showing rainbow colors is identified as Quartz

CHAPTER THREE

3.0 RESULTS AND DISCUSSIONS

Petrography of Metasedimentary rocks

The foliated or banded texture, presence of foliated minerals, and evidence of deformation (like folds and lineation) that were not present in the original sedimentary rock indicate a metamorphosed sedimentary rock (fig. 28). The grain sizes and textures in the metasedimentary rocks help to identify the type of metasedimentary rock. Metasedimentary rocks such as Meta conglomerate, Quartzite, slate etc.

Slate

A low-grade metamorphosed shale that is primary composed of quartz and mica (muscovite and biotite), chlorite and other minerals especially magnetite. Rock also gives slaty cleavages, with a grey/dark color evident that it is biotite rich. Other hand-picked samples with shades of green show their chlorite rich. (fig. 28 A)

Meta-conglomerate

The rounded to sub-rounded grains or clasts within the fine-grained matrix are a texture characteristic of conglomerate, and the Slightly Stretched grains suggest some extent of metamorphism. The thin section analysis suggests the rock is composed of quartz, fragments of quartzite, and other fragments of metavolcanic and metasedimentary rocks derived from the Birimian. The matrix is composed of fine-grained quartz and other altered minerals. (fig. 28. B)

Quartzite

The interlocking quartz crystals, over 90 percent quartz content, hardness and non-foliated nature of the rock indicates the rock is quartzite. The fracture pattern, whereas the rock breaks through the quartz grains but not around the individual sand grains is an indication that the rock has undergone metamorphism. (Fig 28. C)

Schist

Foliated texture, characterized by the alignment of mineral grains. Elongated and somewhat parallel arrangement of the lighter colored minerals suggests foliation. The lighter colored minerals are micas (muscovite and biotite), The darker areas represent other minerals that have undergone alteration. (Fig 28. D)

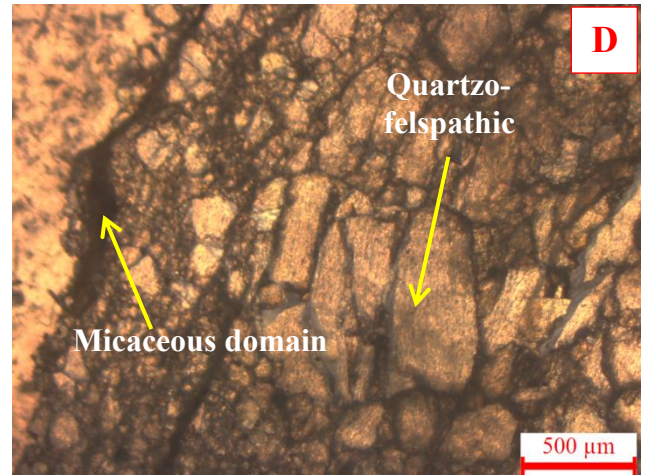
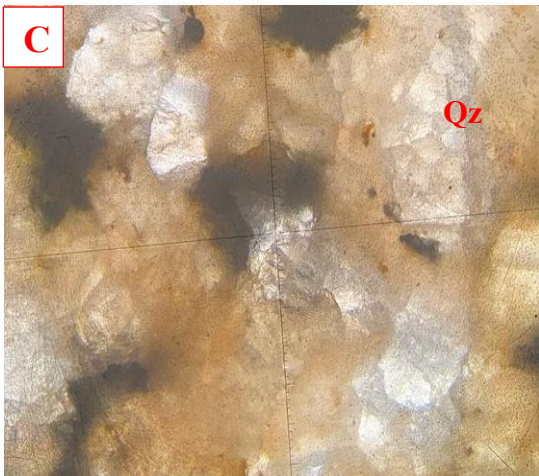
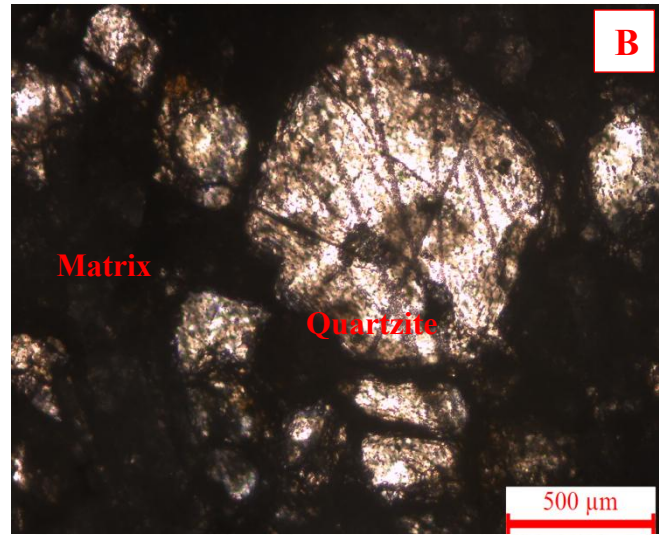
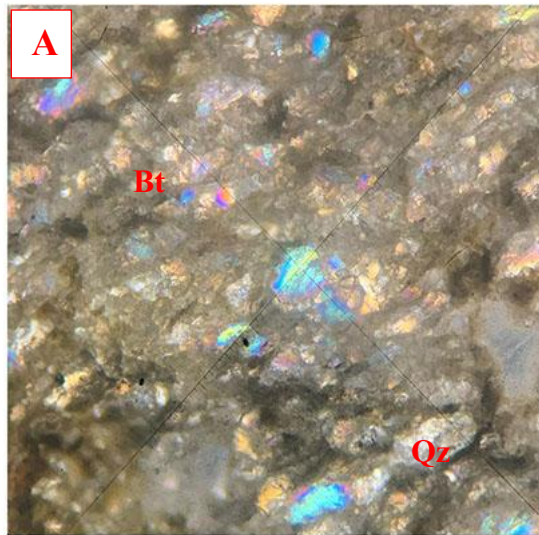


Figure 28. Photomicrograph showing minerals of: (A) slate in XPL, (B) Meta-conglomerate in PPL, (C) Quartzite in PPL, and (D) Schist in PPL.

Petrography of Metavolcanic rocks

The evidence of both volcanic origin and (like lava flow) and subsequent metamorphism (alteration under heat and pressure) resulting in altered textures and some preserved features indicates a metavolcanic rock. The textures and color of the metavolcanic rock show the type of metavolcanic rock.

Metabasalt

Rock exhibits characteristics of heat and pressure history whilst retaining basic chemical composition and mineralogy. Rock is fine grain and contains minerals such as chlorite and other minerals such as plagioclase, quartz and amphiboles.

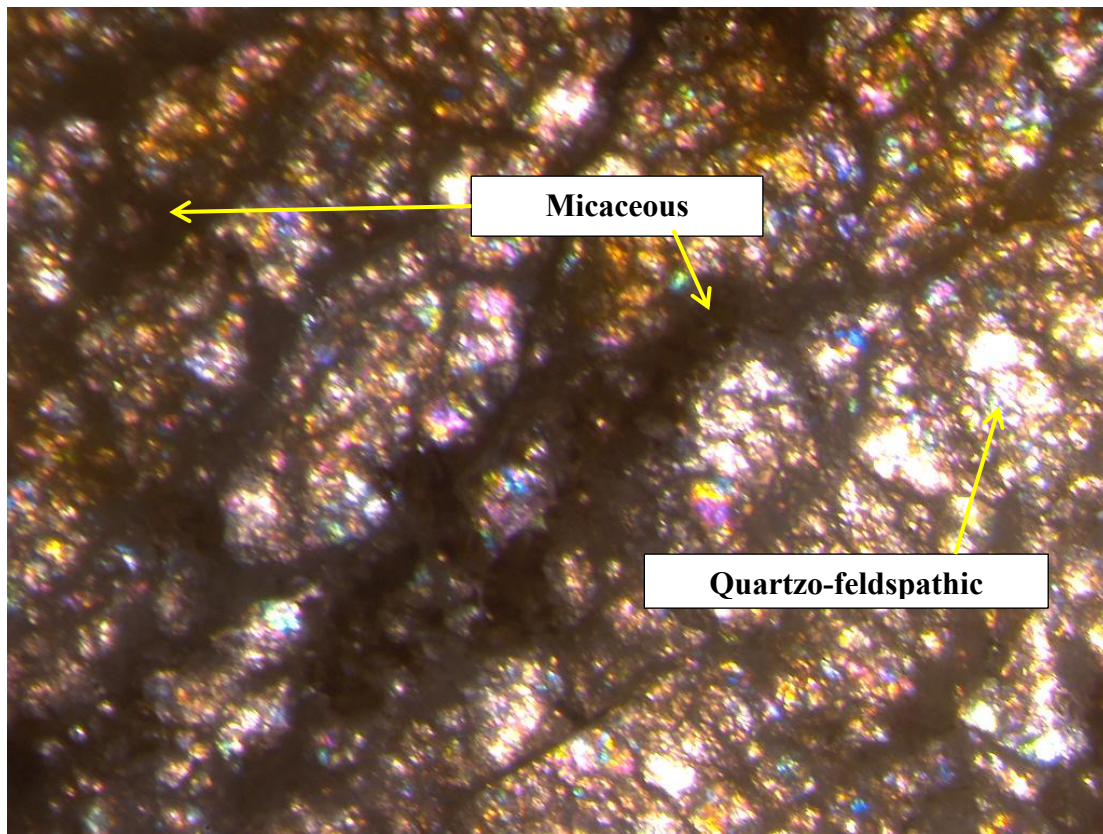


Figure 29 Photomicrograph of Metabasalt in XPL

Petrography of Granitoids

The coarse grains of the igneous rock prove they are intrusive. It also holds 20 to 60 percent quartz and contains other minerals like plagioclase, and alkali feldspar. (orthoclase, sanidine, or microcline).

Granite

The rock is a coarse-grained, with quartz percentage of 45 and 30 percent of feldspar. The remaining percentage is attributed to Micas such as biotite and muscovite, which appear as dark, shiny flakes.

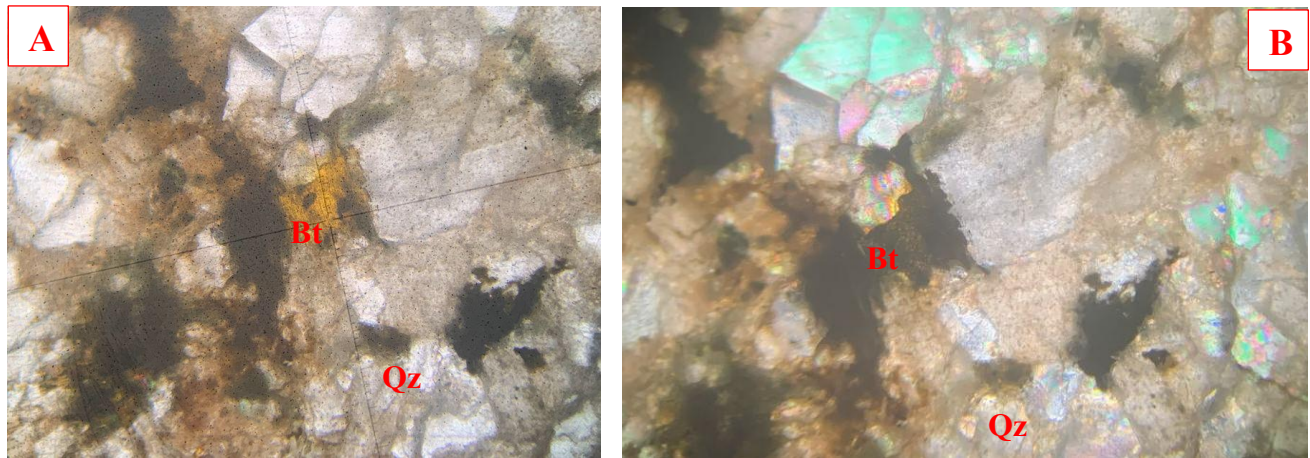


Figure 30 Photomicrograph showing Granite in: (A) PPL, (B) XPL

Structures

Foliations

In figure 31. A, A cleavage (S1) develops along the base of a deforming platy fabrics. A new foliation (S2) places itself perpendicular to the deformation (D1), thereby shortening it in width.

In the same figure, (fig 31. C), Describes a series of deformation histories (D1, D2, D3, D4) causing different foliation directions (S1, S2, S3, S4) in different groups of bands in a slate. The original orientation of first deposited sediment is denoted (S₀) for reference sake.

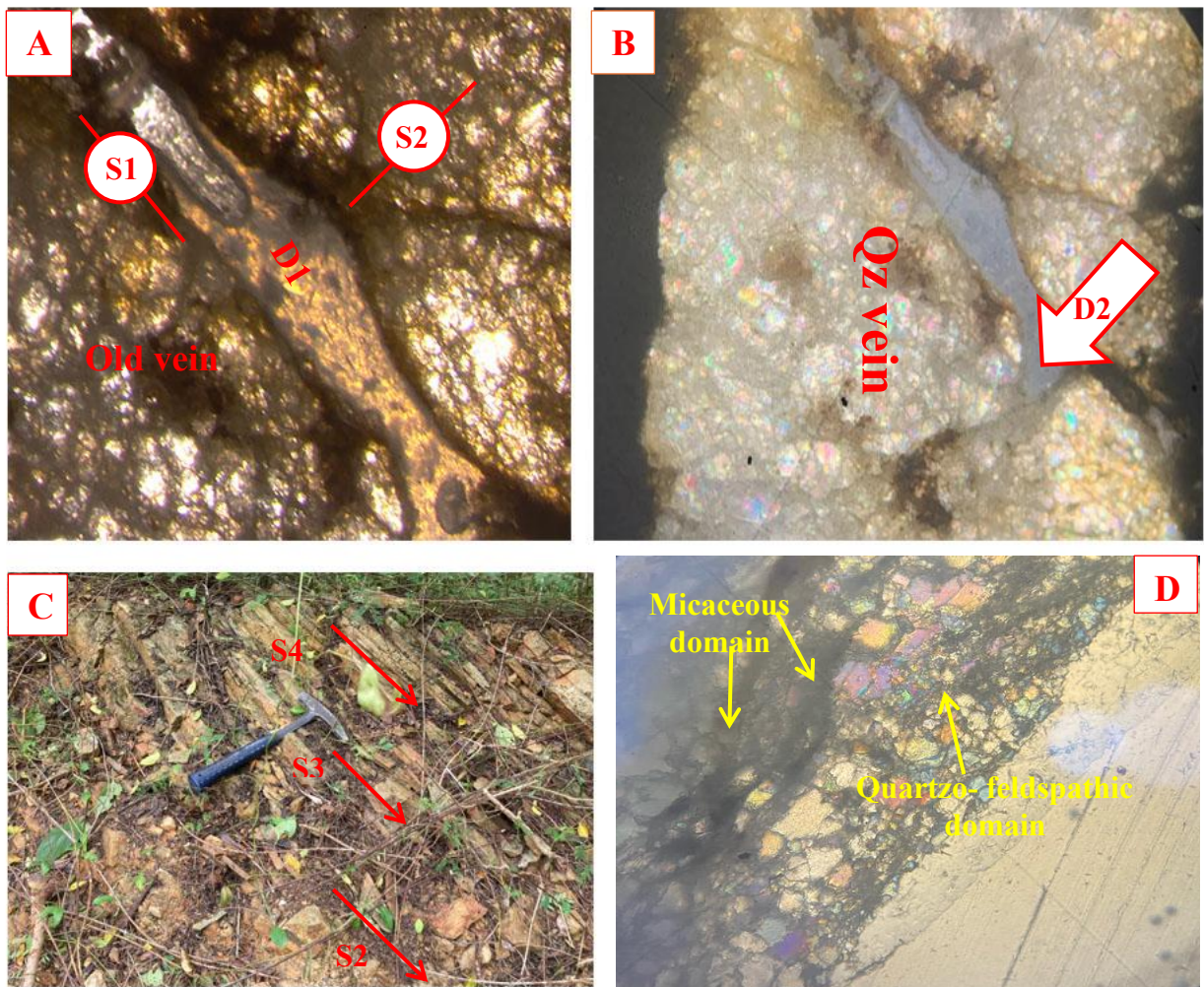


Figure 31 Photomicrograph foliations in: (A) slate in PPL, showing “D1” and “S1” as deformation one and Foliation respectively in a quartz vein, (B) Slate in XPL, (C) series of foliations due to different deformation history in slate, and (D) metamorphosed shale in XPL

Deformation zones

In figure 31.A, The deformation event D1 produces a cleavage trending towards the NE-SW. The second Deformation event is still developing a cleavage across D1 at it's base in the NW-SE direction. (Fig 31.B)

Folds

In figure 32 below, is the evidence of a folding event in a metamorphosed and highly weathered volcanic rock, striking N125° S

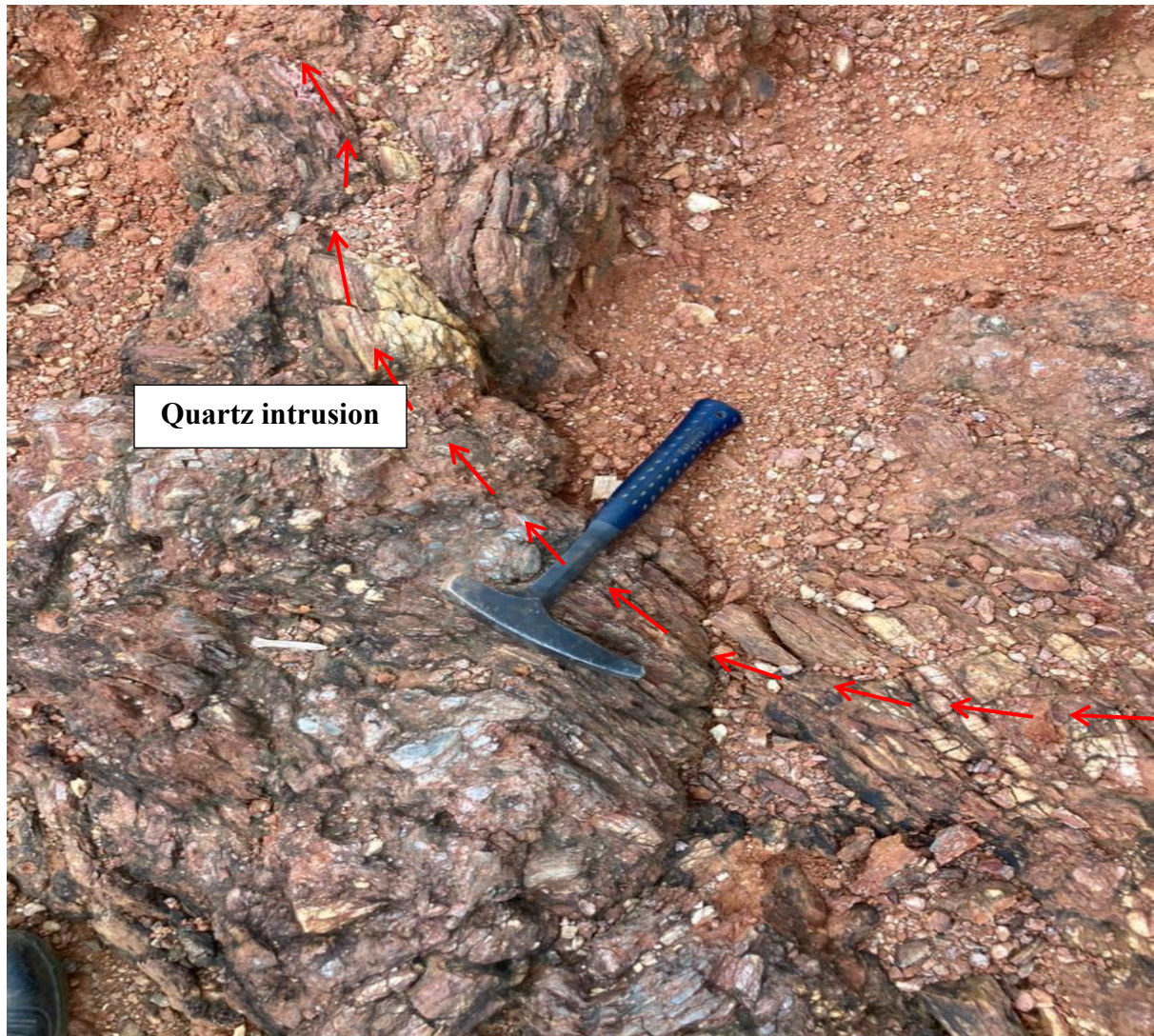


Figure 32 Evidence of folding

CHAPTER FOUR

4.0 CONCLUSION AND RESULTS

Field mapping has provided valuable insights into the diverse rock formations encountered throughout the study area underlain by the Tarkwain and Birimian. The outcrops observed exhibited a range of characteristics, including varying degrees of weathering, grain sizes, textures, and structural features. Key findings include the presence of metamorphic rocks with pronounced foliation, quartz intrusions, and evidence of significant geological processes such as metamorphism and deformation. The variability in rock types, from volcanic and metasedimentary to quartzite, highlights the complex geological history of the region. Overall, this expedition has enhanced our understanding of the local geology, contributing to the broader knowledge of the area's geological evolution.

4.1 APENDICE

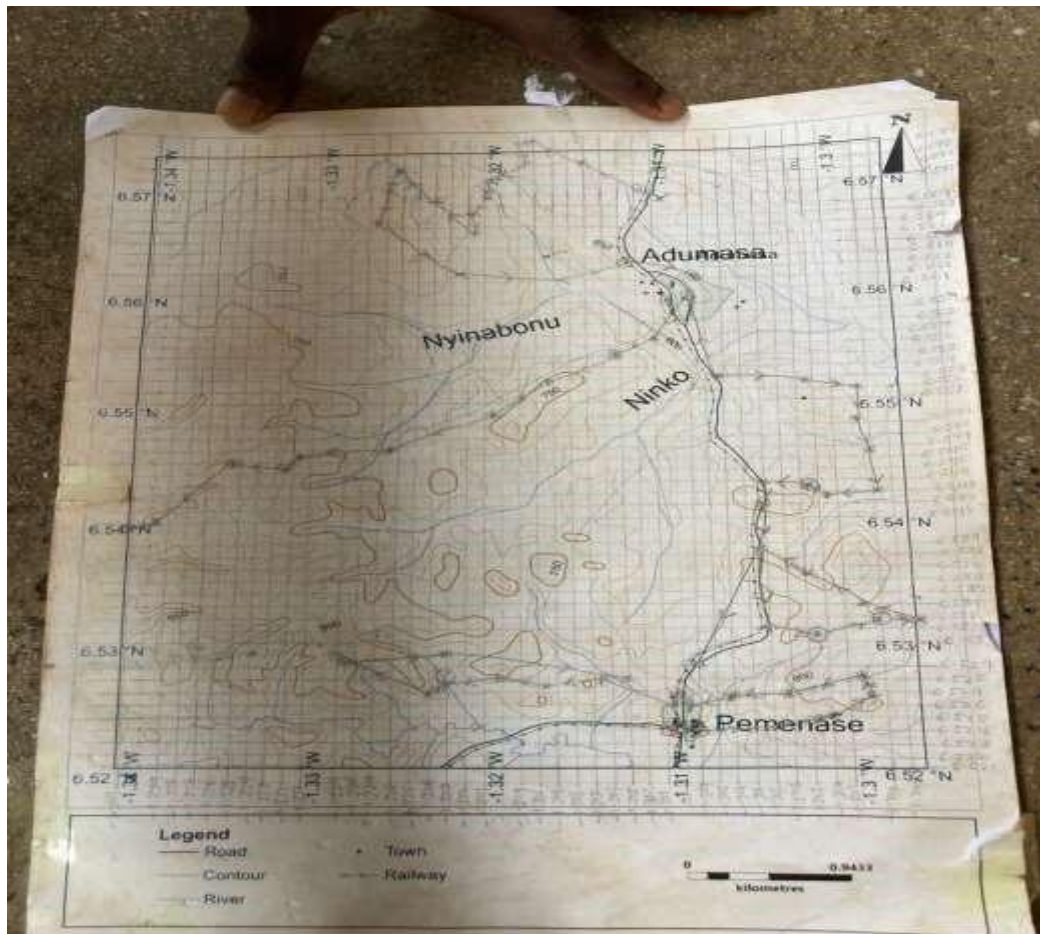


Figure 33 Traverse map used on field

4.1 REFERENCES

www.geologyofghana.com

Principle of igneous and metamorphic petrology by Anthony R. Philpotts, 1990

Harvey Blatt 1982, Petrology: Igneous, sedimentary and metamorphic